Matrix Theory

11 Linear Transformations

Dr. Konda Reddy Mopuri
Dept. of AI, IIT Hyderabad
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Matrices move subspaces

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
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- Matrices are not just a bunch of numbers

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- Matrices are not just a bunch of numbers
- They transform subspaces
- E.g., null space $\rightarrow$ zero vector and all vectors (in domain) $\rightarrow$ column space

Consider an $n \times n$ matrix A

- A transforms x to Ax

$A_{m \times n}$ $\quad n \times 1 \longrightarrow m \times 1$

Consider an $n \times n$ matrix A

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- A transforms x to Ax
- Maps the space onto itself

$A: \mathbb{R}^n \longrightarrow \mathbb{R}^n$

Example-1

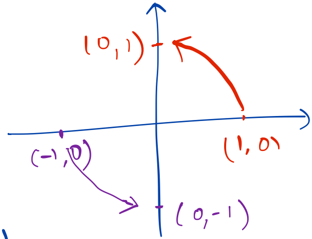
$$A = \begin{bmatrix} c & 0 \\ 0 & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} cx \\ cy \end{bmatrix} = c \begin{bmatrix} x \\ y \end{bmatrix}$$

Scaling or shrinking

$|c| > 1$ $\quad$ $|c| < 1$

Example-2

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -y \\ x \end{bmatrix}$$



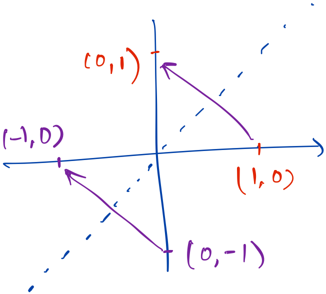
Rotating by 90° counter-clock wise direction

Example-3

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}$$

mirror w.r.t. $y=x$ line

(reflection about the line $y=x$)

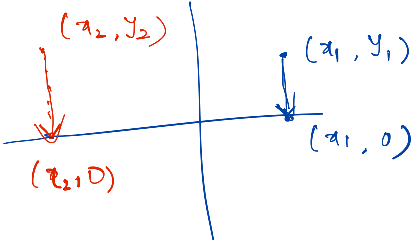


Example-4

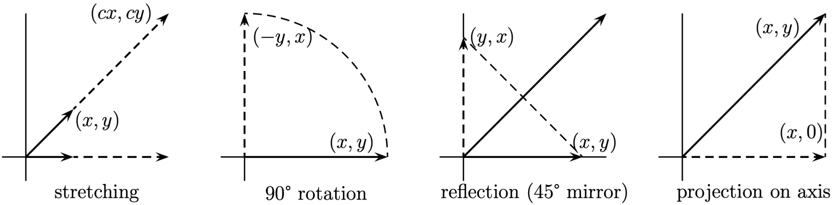
$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 0 \end{bmatrix}$$

projects onto the

x- axis



Example Transformations



Example-5

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ y \end{bmatrix}$$

Streches the xy-plane along the x-axis

Linear Transformation

A transformation T is linear if $T(u + v) = T(u) + T(v)$
and
$T(cu) = cT(u)$.

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- A linear transformation preserves linear combinations.

Example-6

- Translation: $T(x, y) = (x + 1, y)$

$$T((0,0)) = (1,0) \longrightarrow \mathrm{I}$$

$$T(2\cdot(0,0)) = T((0,0)) = (1,0) \longrightarrow \mathrm{II}$$

$$2\cdot T((0,0)) = 2(1,0) = (2,0) \longrightarrow \mathrm{III}$$

$$\mathrm{II} \neq \mathrm{III}$$

Not a linear transformation

Example-7

- Squaring: $T(x, y) = (x^2, y)$

$T((1,0)) = (1,0)$ $\qquad$ $T((2,0)) = (4,0)$

$T((1,0)+(2,0)) = T((3,0)) = (9,0) \longrightarrow \text{I}$

$T((1,0)) + T((2,0)) = (1,0) + (4,0)$

$= (5,0) \longrightarrow \text{II}$

$\text{I} \neq \text{II}$ $\qquad$ Not a linear transformation

Matrix $\rightarrow$ Linear Transformation

- Does a matrix multiplication always define a linear transformation?

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- For any matrix $A \in \mathbb{R}^{m \times n}$, define $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ as $T(x) = Ax$

Matrix $\to$ Linear Transformation

- Does a matrix multiplication always define a linear transformation?
- For any matrix $A \in \mathbb{R}^{m \times n}$, define $T : \mathbb{R}^n \to \mathbb{R}^m$ as $T(x) = Ax$
- $T(\alpha u + \beta v) = A(\alpha u + \beta v) = \alpha Au + \beta Av$ $= \alpha T(u) + \beta T(v)$ ✓

α, β are Scalars

Does every linear transformation lead to a matrix?

- Yes, once the bases are chosen

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- Follows nicely from the standard basis

Consider a linear transformation $T(\cdot)$

Operate it on a vector $x \Rightarrow T(x)$

but $x = \sum_{i=1}^{n} x_i \cdot e_i$ $\qquad$ e_i - standard basis

Since T is linear, $T(x) = T\left(\sum_{i=1}^{n} x_i e_i\right) = \sum_{i=1}^{n} x_i \cdot T(e_i)$

this can be represented as

$$\begin{bmatrix} | & | & & | \\ T(e_1) & T(e_2) & \cdots & T(e_n) \\ | & | & & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = Ax$$

Does every linear transformation lead to a matrix?

- Yes, once the bases are chosen
- Follows nicely from the standard basis

To know a linear transformation completely, it is enough to know what it does to a basis.

Example - Polynomials

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- Basis for P_3: $\{p_1 = 1, p_2 = t, p_3 = t^2, p_4 = t^3\}$ (not unique!)

Example - Polynomials

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॥ नवोन्मेषेण देश निर्माणम् ॥

- Basis for P_3: $\{p_1 = 1, p_2 = t, p_3 = t^2, p_4 = t^3\}$ (not unique!)
- Consider the transformation: Derivative $(\frac{\partial}{\partial t})$

$\frac{\partial}{\partial t} p_1 = 0$ $\quad \frac{\partial}{\partial t} p_2 = 1$ $\quad \frac{\partial}{\partial t} p_3 = 2t$ $\quad \frac{\partial}{\partial t} p_4 = 3t^2$

$p_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ $\quad p_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ $\quad p_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ $\quad p_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$ $\quad \mathbb{R}^4$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{4\times 4}$$

$AP_1 \quad AP_2 \quad AP_3 \quad AP_4$

$N(A) = c\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ $\qquad C(A) =$

Example - Polynomials

$$\underset{\text{diff}}{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \underbrace{\begin{bmatrix} 2 \\ 1 \\ -1 \\ -1 \end{bmatrix}}_{x} = \begin{bmatrix} 1 \\ 2 \\ -3 \\ 0 \end{bmatrix}$$

$$p = 2 + t - t^2 - t^3$$

$$1 - 2t - 3t^2 + 0t^3$$

$$= 1 \cdot P_1 - 2P_2 - 3P_3 + 0 \cdot P_4$$

$A_{4\times 4}$

$r = 3$

dim. of $N(A) = 1$

$$N(A) = c\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$C(A) = a\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + b\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\frac{\partial}{\partial t} : \mathbb{P}^3 \to \mathbb{P}^3$$

Transformation of a space to itself

- Matrix carries all the essential information.
- If the basis is known, and the matrix is known, then the transformation of every vector is known

Transformation of a space to itself

- Matrix carries all the essential information.
- If the basis is known, and the matrix is known, then the transformation of every vector is known
- A transformation from one space to another requires a basis for each.

Transformation between two subspaces

2U Suppose the vectors $x_1, \ldots, x_n$ are a basis for the space $\mathbf{V}$, and vectors $y_1, \ldots, y_m$ are a basis for $\mathbf{W}$. Each linear transformation T from $\mathbf{V}$ to $\mathbf{W}$ is represented by a matrix A. The jth column is found by applying T to the jth basis vector x_j, and writing $T(x_j)$ as a combination of the y's:

$$\textbf{Column } j \textbf{ of } A \qquad T(x_j) = Ax_j = a_{1j}y_1 + a_{2j}y_2 + \cdots + a_{mj}y_m. \qquad (6)$$

$T: V \longrightarrow W$

$\{x_i\} \qquad \{y_i\}$

$A = \begin{bmatrix} | & | & & | \\ Ax_1 & Ax_2 & \cdots & Ax_n \\ | & | & & | \end{bmatrix}$

Example - Polynomials

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Example - Polynomials

- Basis for P_3: $\{p_1 = 1, p_2 = t, p_3 = t^2, p_4 = t^3\}$ (not unique!)
- Consider the transformation: Integration $(\int_0^t)$: $\mathbb{P}^3 \longrightarrow \mathbb{P}^4$

$x_1 = 1$ $\quad x_2 = t$ $\quad x_3 = t^2$ $\quad x_4 = t^3$

$Ax_1 = t$ $\quad Ax_2 = t^2/2$ $\quad Ax_3 = t^3/3$ $\quad Ax_4 = t^4/4$

$y_1 \ \ 1$

$y_2 \ \ t$

$y_3 \ \ t^2$

$y_4 \ \ t^3$

$y_5 \ \ t^4$

$$A_{int} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & y_2 & 0 & 0 \\ 0 & 0 & y_3 & 0 \\ 0 & 0 & 0 & y_4 \end{bmatrix} \quad 5 \times 4$$

$N(A) = \{0\}$ $\quad r = 4$

$$\begin{matrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{matrix}$$

Example - Polynomials

- Basis for P_3: $\{p_1 = 1, p_2 = t, p_3 = t^2, p_4 = t^3\}$ (not unique!)
- Consider the transformation: Integration $(\int_0^t)$
- Transformation from $V = P_3$ to $W = P_4$

$\underline{A_{diff}}\, A_{int} \, (\cdot) = (\cdot)$

5×4

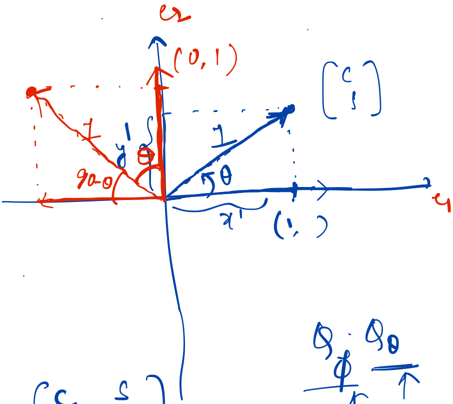
Rotation (Q)

$\mathbb{R}^2$

T: c.c by θ

$$\begin{bmatrix} | & | \\ Ae_1 & Ae_2 \\ | & | \end{bmatrix}$$

$$A = \begin{bmatrix} c & -s \\ s & c \end{bmatrix}$$



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Q_θ

$$Q_{-\theta} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix}$$

$$Q_\theta \cdot Q_{-\theta} = \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} c & s \\ -s & c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

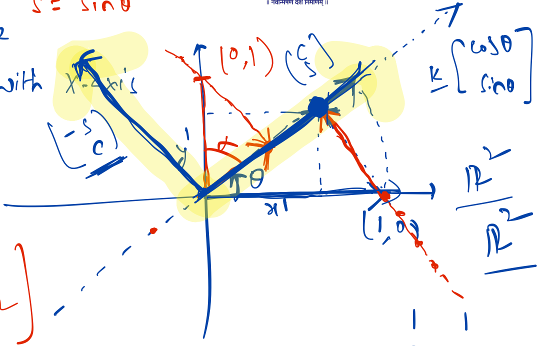
$$Q_\phi \cdot Q_\theta = Q_{(\theta+\phi)}$$

Projection (P)

$c = \cos\theta$

$s = \sin\theta$

T: projecting onto a line that makes an angle θ with x-axis



$$\begin{bmatrix} | & | \\ Ae_1 & Ae_2 \\ | & | \end{bmatrix}$$

$$P = \begin{bmatrix} c\begin{bmatrix} c \\ s \end{bmatrix} & s\begin{bmatrix} c \\ s \end{bmatrix} \end{bmatrix} = \begin{bmatrix} c^2 & sc \\ cs & s^2 \end{bmatrix}$$

$\mathbb{R}^2 \quad \mathbb{R}^2$

$$\begin{bmatrix} | & | \\ e_1 & e_2 \\ | & | \end{bmatrix}$$

Invertible ✗

Ae_1

$$\begin{bmatrix} c^2 \\ cs \end{bmatrix} \cancel{=} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$P(Px) = Px$$

$$P^2 = P$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Reflection (H)

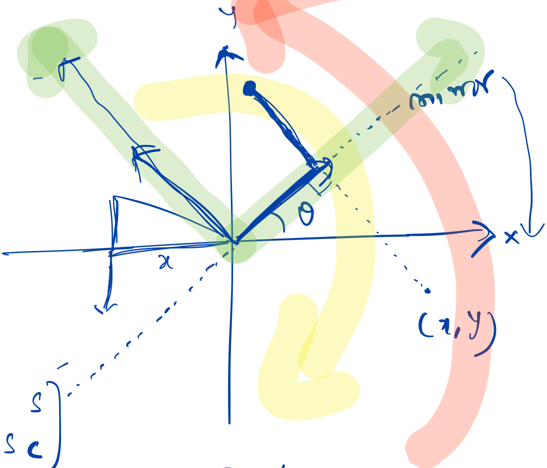
$$\begin{bmatrix} | & | \\ Ae_1 & Ae_2 \\ | & | \end{bmatrix}$$

$$Ref_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$H = Rot_{\theta}\, Ref_x\, Rot_{-\theta}$$

$$= \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} c & s \\ -s & c \end{bmatrix}$$

$$= \begin{bmatrix} c & -s \\ s & c \end{bmatrix} \begin{bmatrix} c & s \\ s & -c \end{bmatrix} = \begin{bmatrix} c^2 - s^2 & 2sc \\ 2sc & s^2 - c^2 \end{bmatrix} = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$$



$Ae_1 = 1 \cdot y_1 + 0 \cdot y_2$

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Summary

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A matrix is not just a bunch of numbers. It represents a transformation of space. It becomes an operator acting on vectors. $T(x) = Ax$.

A linear transformation preserves linear combinations. $\Rightarrow T(0) = 0$.

Basis vectors determine everything.

2x

Next

- Orthogonality

Working Space-1

-

Working Space-2

-

Working Space-3

-